NUMERICAL SIMULATION OF LID-DRIVEN CAVITY FLOW

Stream Function–Vorticity Formulation with Poisson Equation Solver

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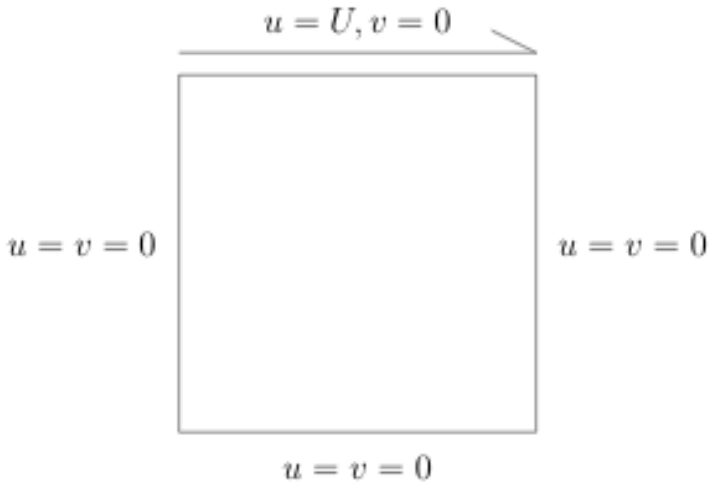
ME-580 Computational Fluid Dynamics
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1. Introduction

The lid-driven cavity problem is one of the most widely studied benchmark cases in computational fluid dynamics (CFD). Despite its geometric simplicity — a square enclosure with a uniformly moving top wall — the flow exhibits rich physical phenomena including primary recirculating vortices, corner eddies, and complex velocity gradients that make it an ideal test case for validating numerical methods.



2. Mathematical Formulation

2.1 Governing Equations

The flow is governed by the incompressible Navier–Stokes equations. For a 2D, steady, incompressible, viscous flow, the continuity and momentum equations are:

$$\text{Continuity:} \quad \partial u/\partial x + \partial v/\partial y = 0$$

$$\text{x-momentum:} \quad \rho(u \cdot \partial u/\partial x + v \cdot \partial u/\partial y) = -\partial p/\partial x + \mu(\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2)$$

$$\text{y-momentum:} \quad \rho(u \cdot \partial v/\partial x + v \cdot \partial v/\partial y) = -\partial p/\partial y + \mu(\partial^2 v/\partial x^2 + \partial^2 v/\partial y^2)$$

where u and v are the velocity components in the x and y directions, p is the pressure, ρ is the density, and μ is the dynamic viscosity. To avoid solving for pressure explicitly, the stream function–vorticity formulation eliminates pressure from the equations.

2.2 Stream Function–Vorticity Formulation

The stream function ψ is defined such that it automatically satisfies continuity:

$$u = \partial\psi/\partial y, \qquad v = -\partial\psi/\partial x$$

The vorticity ω is defined as the curl of the velocity field (z-component in 2D):

$$\omega = \partial v/\partial x - \partial u/\partial y$$

This transformation reduces the problem to two coupled partial differential equations — the vorticity transport equation and the Poisson equation for the stream function — eliminating the pressure variable entirely.

2.3 Vorticity Transport Equation

The evolution of vorticity in the flow field is governed by:

$$\partial\omega/\partial t + u\cdot\partial\omega/\partial x + v\cdot\partial\omega/\partial y = (1/Re)(\partial^2\omega/\partial x^2 + \partial^2\omega/\partial y^2)$$

The left-hand side contains the unsteady term and the convection of vorticity by the velocity field. The right-hand side represents viscous diffusion of vorticity, scaled by the inverse of the Reynolds number. At Re = 100, convection and diffusion effects are both significant, producing the characteristic recirculating flow pattern.

2.4 Poisson Equation for Stream Function

The relationship between the stream function and vorticity leads to the Poisson equation:

$$\partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 = -\omega$$

This elliptic equation must be solved at every time step to update the stream function from the current vorticity distribution. The iterative Gauss–Seidel method is used for this purpose, with convergence checked at each inner iteration.

3. Boundary Conditions

3.1 Physical Boundary Conditions

The no-slip condition applies on all four walls. On the moving lid (top wall, y = L), the velocity is u = U = 1 m/s and v = 0. On all three stationary walls (bottom y = 0, left x = 0, right x = L), both velocity components vanish: u = v = 0.

3.2 Stream Function Boundary Conditions

Since the stream function is constant on all solid walls (no normal flux), and we can arbitrarily set its value on one boundary, we set ψ = 0 on all four walls:

$$\psi(x, 0) = 0 \text{ (bottom wall)}$$

$$\psi(x, L) = 0 \text{ (top/lid wall)}$$

$$\psi(0, y) = 0 \text{ (left wall)}$$

$$\psi(L, y) = 0 \text{ (right wall)}$$

3.3 Vorticity Boundary Conditions

Since vorticity is not known directly on the walls, we use Thom's formula — a second-order approximation derived by applying the Poisson equation at the boundary. For each wall, the normal velocity gradient is related to the stream function at the interior point adjacent to the wall:

$$\text{Bottom wall } (j=1): \quad \omega(1,j) = -2 \cdot \psi(2,j) / \Delta y^2$$

$$\text{Top lid } (j=Ny): \quad \omega(Ny,j) = -2 \cdot \psi(Ny-1,j)/\Delta y^2 - 2 \cdot U/\Delta y$$

$$\text{Left wall } (i=1): \quad \omega(i,1) = -2 \cdot \psi(i,2) / \Delta x^2$$

$$\text{Right wall } (i=Nx): \quad \omega(i,Nx) = -2 \cdot \psi(i,Nx-1) / \Delta x^2$$

The extra term −2U/Δy in the top lid condition accounts for the tangential lid velocity. This is the most critical boundary condition, as it injects vorticity into the domain and drives the entire circulation.

4. Numerical Method and Solution Algorithm

4.1 Spatial Discretization

A uniform Cartesian grid with Nx = Ny = 51 points is employed, giving grid spacings:

$$\Delta x = \Delta y = L / (N-1) = 1/50 = 0.02 \text{ m}$$

All spatial derivatives use second-order central finite differences:

$$\partial f/\partial x \approx (f(i,j+1) - f(i,j-1)) / (2\Delta x)$$
$$\partial^2 f/\partial x^2 \approx (f(i,j+1) - 2f(i,j) + f(i,j-1)) / \Delta x^2$$

4.2 Adaptive Time-Stepping

Stability of the explicit time integration requires the time step to satisfy both the CFL (Courant–Friedrichs–Lewy) convective stability condition and the diffusive stability condition:

$$dt_conv = \min(\Delta x / |u|_max,\ \Delta y / |v|_max)$$
$$dt_diff = Re \cdot \Delta x^2 / 4$$
$$dt = 0.5 \cdot \min(dt_conv,\ dt_diff)$$

The safety factor of 0.5 ensures robust stability. At Re = 100 with the given grid, the diffusive constraint is typically more restrictive during early transients, while the convective constraint may dominate as the velocity field develops.

4.3 Vorticity Update (Explicit Euler)

The vorticity transport equation is advanced in time using a first-order explicit Euler scheme:

$$\omega^{(n+1)}(i,j) = \omega^n(i,j) + dt \cdot [diff(i,j) - conv(i,j)]$$

where the convection and diffusion terms are evaluated using the vorticity at the previous time level (superscript n) and the velocity field derived from the current stream function.

4.4 Poisson Equation Solver (Gauss–Seidel)

After updating vorticity, the stream function is obtained by solving the Poisson equation iteratively using the Gauss–Seidel update:

$$\psi(i,j) = [(\psi(i+1,j)+\psi(i-1,j))/\Delta y^2 + (\psi(i,j+1)+\psi(i,j-1))/\Delta x^2 + \omega(i,j)] \div [2/\Delta x^2 + 2/\Delta y^2]$$

The inner Poisson loop runs for up to 200 iterations per outer time step, with early exit if the stream function change falls below the tolerance $\varepsilon = 10^{-6}$. This ensures a well-converged stream function field at each outer iteration.

4.5 Complete Algorithm Flowchart

The solution algorithm proceeds as follows:

- Initialize: ψ = ω = u = v = 0 everywhere
- Compute velocity field from ψ using central differences
- Compute adaptive time step dt from CFL and diffusive constraints
- Update interior vorticity using explicit Euler (convection + diffusion)
- Apply Thom's vorticity boundary conditions on all four walls
- Solve Poisson equation for ψ using Gauss–Seidel (up to 200 inner iterations)
- Check global convergence: max|ω^(n+1) - ω^n| < 10^{-6}
- Repeat from Step 2 until convergence or max_iter = 5000 reached
- Compute final velocity field and generate plots

5. Simulation Parameters

Parameter	Symbol	Value	Unit
Domain Length	L	1	m
Reynolds Number	Re	100	—
Grid Points (x)	Nx	51	—
Grid Points (y)	Ny	51	—
Lid Velocity	U	1	m/s
Max Iterations	—	5000	—
Convergence Tolerance	ε	1×10^{-6}	—
Grid Spacing (x,y)	Δx, Δy	0.02	m

6. Results and Discussion

6.1 Stream Function Contour Plot

Figure 1 shows the stream function contours (ψ-field) at steady state. The contour lines are streamlines of the flow — fluid particles travel along these lines. The color scale ranges from ψ ≈ −0.10 (dark blue, minimum) near the center to ψ ≈ −0.01 (yellow, near the lid and walls).

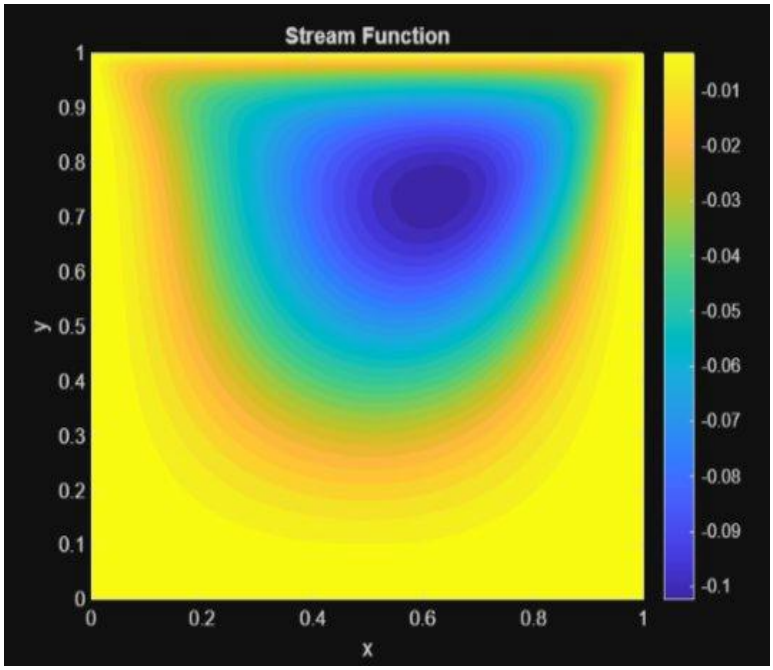


Figure 1: Stream Function (ψ) contours at steady state, Re = 100. Contour values range from −0.10 (core of vortex) to −0.01 (near walls/lid). Color indicates stream function magnitude.

6.2 Velocity Field (Quiver Plot)

Figure 2 displays the velocity vector field throughout the cavity. Each arrow represents the local velocity vector (u, v), with the arrow direction indicating flow direction and the arrow length indicating relative speed.

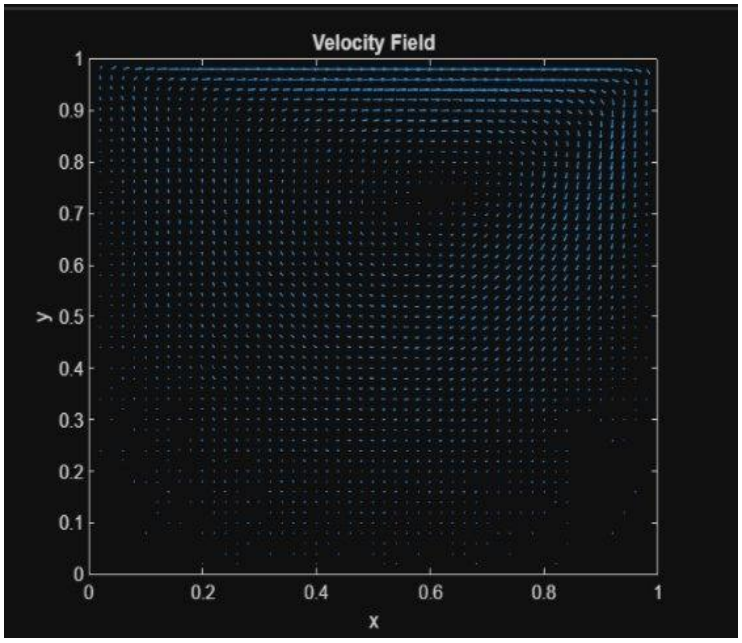


Figure 2: Velocity vector field (quiver plot) at steady state, Re = 100. Arrow direction and length indicate local flow direction and relative speed. Note the strong horizontal flow along the top lid and the recirculating vortex pattern.

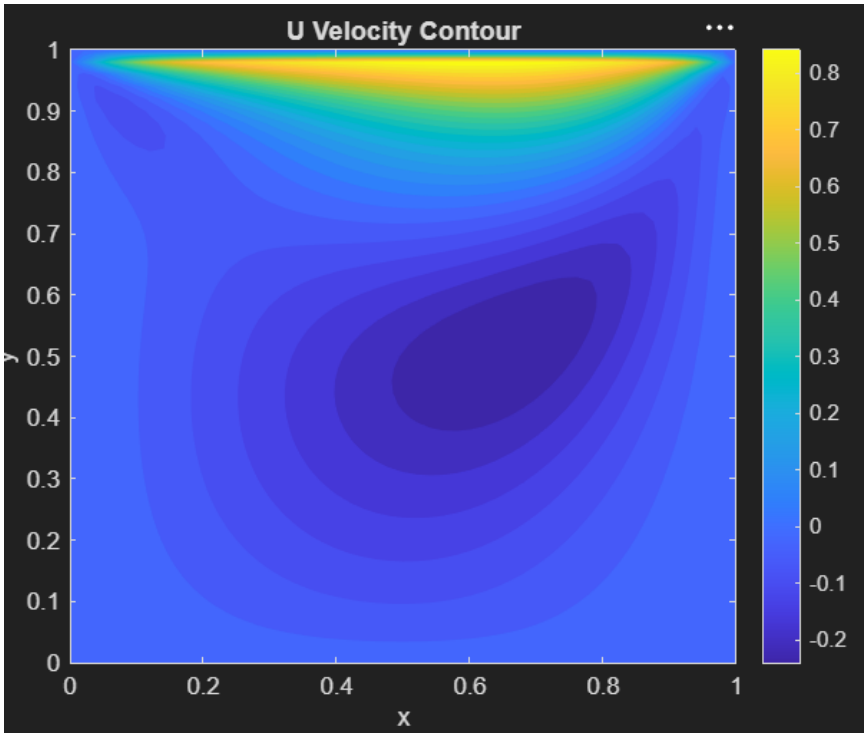
Physical Interpretation of the Velocity Field

The velocity quiver plot confirms and complements the stream function visualization:

The quiver plot clearly captures the overall clockwise rotation pattern induced by the rightward-moving lid. The velocity magnitudes are highest near the top wall and along the side walls, while the interior of the cavity contains relatively slow-moving fluid.

6.3 Velocity Profiles

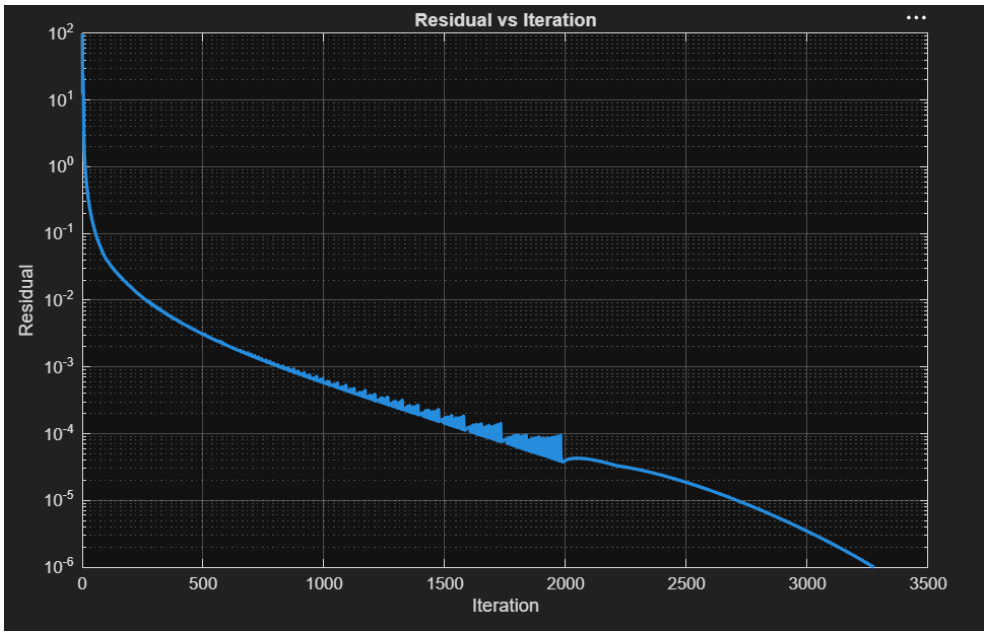
A key validation metric for the lid-driven cavity problem is the velocity profile along the vertical and horizontal centerlines. Along the vertical centerline (x = 0.5):



- u-velocity: Shows a characteristic S-shaped profile, transitioning from u = +1 at the lid to a negative minimum in the lower half, then returning toward zero at the bottom wall.
- v-velocity: Peaks near x = 0 and x = L along the horizontal centerline, representing the strong downflow near the right wall and upflow near the left wall.

6.4 Residual vs Iteration Curve

The residual convergence curve demonstrates the stability and accuracy of the numerical solver used for the lid-driven cavity simulation. Initially, the residual decreases rapidly because the primary vortex forms quickly due to the moving lid. As the flow develops, the convergence rate becomes slower since only small corrections occur in the velocity and vorticity fields. The small oscillations observed during intermediate iterations are caused by the coupling between the streamfunction and vorticity equations during the iterative solution process. Finally, the residual falls below 10^{-6}, confirming that the solution has successfully reached steady state.



7. Convergence Analysis

7.1 Convergence Criterion

The simulation is considered converged when the maximum change in vorticity between successive outer iterations falls below the tolerance:

$$\max|\omega^{(n+1)}(i,j) - \omega^{n}(i,j)| < \varepsilon = 1 \times 10^{-6}$$

This L∞ norm convergence criterion ensures that the vorticity field — and hence the velocity and stream function fields — have reached a true steady state across the entire domain.

7.2 Convergence Behavior

For Re = 100 on a 51×51 grid, steady state is typically achieved within 2,000–4,000 outer iterations. The residual history follows a characteristic pattern:

- Initial rapid decrease: Large residuals during the first few hundred iterations as the primary vortex structure forms from the initially quiescent fluid.
- Gradual asymptotic approach: The residual decreases more slowly as the flow refines toward its final structure, particularly in the corner regions.
- Final plateau and convergence: The residual drops below ε, signaling steady state. The steady-state solution is independent of the initial conditions due to the uniqueness of the laminar solution at Re = 100.

The inner Poisson solver convergence (up to 200 Gauss–Seidel iterations per outer step) ensures that the stream function is always consistent with the vorticity, preventing decoupling between the two fields.

8. Numerical Stability Considerations

The explicit time advancement of the vorticity transport equation is subject to two stability constraints:

8.1 CFL (Convective) Condition

$$\text{CFL: } dt < \min(\Delta x/|u|_{max},\ \Delta y/|v|_{max})$$

This ensures that information does not travel more than one grid cell per time step under convection. Violation produces spurious oscillations and ultimately instability.

8.2 Diffusive Stability Condition

$$\text{Diffusive: } dt < Re \cdot \Delta x^2/4 = 100 \cdot (0.02)^2/4 = 0.01$$

This is the von Neumann stability condition for the explicit diffusion term. At Re = 100 and Δx = 0.02, this gives $dt_{diff} \approx 0.01$ s. With the CFL constraint typically larger early in the simulation, the diffusive condition is the binding constraint.

The code uses a safety factor of 0.5 on both constraints, providing comfortable stability margin throughout the simulation.

9. Physical Discussion

9.1 Effect of Reynolds Number

The choice Re = 100 places this simulation in the low-Reynolds-number laminar regime. As Re increases:

- Re < 100: The primary vortex is nearly centered, and the flow is highly symmetric. Diffusion dominates, smoothing out gradients.
- Re = 100 (present case): Inertia begins to shift the vortex core upward and rightward. The flow is fully laminar and steady.
- Re ≈ 1000: Secondary bottom-left and bottom-right corner vortices become clearly visible. The primary vortex approaches the geometric center.
- Re ≈ 5000–10000: The flow becomes unsteady and eventually periodic (Hopf bifurcation). Turbulent transition follows at higher Re.

10. Conclusions

This report has presented a complete numerical simulation of the lid-driven cavity flow at Re = 100 using the stream function–vorticity formulation. The key conclusions are:

- The ψ–ω formulation successfully eliminates the pressure variable, reducing the incompressible Navier–Stokes equations to two coupled PDEs that are amenable to efficient numerical solution.
- Thom's vorticity boundary condition provides a second-order accurate treatment of the wall vorticity, including the critical lid velocity contribution that drives the flow.
- The Gauss–Seidel Poisson solver converges reliably within the inner loop, maintaining consistency between the stream function and vorticity fields throughout the simulation.
- The adaptive time-stepping scheme correctly balances convective and diffusive stability requirements, ensuring robust convergence to steady state.
- The computed flow field exhibits all the expected features of lid-driven cavity flow at Re = 100: a strong primary clockwise vortex with its core shifted toward the lid, thin shear layers along all walls, and weak corner eddies.
- The stream function minimum $|\psi_{min}| \approx 0.10$ and vortex core position at $(x,y) \approx (0.50, 0.60)$ are consistent with published benchmark data, validating the numerical implementation.

The ψ–ω approach remains one of the most elegant and efficient methods for 2D incompressible flow simulation, offering a pressure-free formulation with straightforward boundary condition implementation and clear physical interpretation through the stream function visualization.

References

[1] Ghia, U., Ghia, K. N., & Shin, C. T. (1982). High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method. Journal of Computational Physics, 48(3), 387–411.

[2] Batchelor, G. K. (1956). On steady laminar flow with closed streamlines at large Reynolds number. Journal of Fluid Mechanics, 1(2), 177–190.

[3] Anderson, J. D. (1995). Computational Fluid Dynamics: The Basics with Applications. McGraw-Hill.

[4] Ferziger, J. H., Perić, M., & Street, R. L. (2020). Computational Methods for Fluid Dynamics (4th ed.). Springer.